# Priority of operations with positive and negative integers, decimals and fractions

**1** Work out $\frac{9}{10} + \frac{3}{5} \times \frac{1}{2}$. Tick **1** correct answer

- ☐ $\frac{4}{3}$
- ☐ $\frac{3}{4}$
- ☐ $\frac{5}{6}$
- ☑ $\frac{6}{5}$

**2** Match the calculations on the left with the correct answer on the right. Write the correct letter in each box

| | |
|---|---|
| a | $\left(\frac{2}{5}\right)^2$ |
| b | $\left(\frac{3}{5}\right)^2$ |
| c | $\left(\frac{4}{3}\right)^2$ |
| d | $\left(\frac{5}{6}\right)^2$ |

| | |
|---|---|
| a | $\frac{4}{25}$ |
| b | $\frac{9}{25}$ |
| d | $\frac{25}{36}$ |
| c | $\frac{16}{9}$ |

3 Select the correct explanation for why the following is correct. Tick **1** correct answer

$$\frac{3}{7}+\frac{1}{2}\times\frac{3}{5}\div\frac{3}{5}=\frac{3}{7}+\frac{1}{2}\times\frac{3}{5}\times\frac{5}{3}$$

- [ ] Using the priority of operations, multiplication comes first before division.
- [ ] Changing fractions to their reciprocals makes things easier.
- [ ] Addition comes after multiplication.
- [x] When dividing, division can be written as multiplication by the reciprocal. ✓

4 Which of the following equals $\frac{16}{25}$? Tick **4** correct answers

- [x] $\frac{4^2}{5^2}$ ✓
- [x] $(\frac{4}{5})^2$ ✓
- [x] $\sqrt{\frac{256}{625}}$ ✓
- [x] $\frac{4}{5}\div\frac{5}{4}$ ✓
- [ ] $\frac{4}{5}\times 2$

**5**

Work out $1 + \sqrt[3]{15\frac{5}{8}} \times \frac{1}{2}$. Tick **1** correct answer

- [x] $\frac{9}{4}$
- [ ] $\frac{4}{9}$
- [ ] 1.5
- [ ] 1
- [ ] $\frac{13}{8}$

**6**

Without using a calculator, work out $3 + \frac{2}{3} - \sqrt{\frac{25}{36}}$. Tick **1** correct answer

- [ ] $1\frac{5}{6}$
- [x] $2\frac{5}{6}$
- [ ] $3\frac{5}{36}$
- [ ] $3\frac{25}{36}$